Python Fundamentals

Practice Question Sheet

Topics: Variables • Data Types • Input/Output • Operators

Name: ____________________ **Date:** ____________________

__________ _________ **Total Questions:** 20

01 Section 1: Variables & Data Types

Q1. *[Output Tracing]*

What will be the output of the following code?

```
x = 10
y = 3.5
z = True
name = 'Python'
print(type(x), type(y), type(z), type(name))
```

Answer:

Q2. *[Fill in the Blank]*

Fill in the blanks to declare the correct data types:

a) A variable storing the value **42** is of type _____________

b) A variable storing **3.14** is of type _____________

c) A variable storing **True** is of type _____________

d) A variable storing **'Hello'** is of type _____________

Answer:

Q3. *[Error Detection]*

Identify and fix the error in the code below:

```
1name = 'Alice'
print(1name)
```

Error & Fix:

Q4. *[Output Tracing]*

What will be the output of the following code? Explain why.

```
a = 5
b = a
a = 100
print(a)
print(b)
```

Answer:

Q5. *[Multiple Choice]*

Which of the following is a valid Python variable name?

A) 2score

B) my-score

C) my_score

D) class

Circle your answer: _________

Answer:

02 Section 2: Input & Output

Q6. *[Code Writing]*

Write a Python program that:

1. Asks the user to enter their name.

2. Asks the user to enter their age.

3. Prints: Hello, [name]! You are [age] years old.

Your Code:

Q7. *[Output Tracing]*

What will be printed if the user enters **25** at the prompt?

```
age = input('Enter your age: ')
print('Type of age:', type(age))
print('Your age is:', age)
```

Answer:

Q8. *[Error Detection]*

The following code crashes. Identify the error and correct it:

```
num = input('Enter a number: ')
result = num + 10
print(result)
```

Error & Fix:

Q9. *[Output Tracing]*

Predict the exact output of this code:

```
x = 7
y = 2
print('Sum     =', x + y)
print('Product =', x * y)
print(f'Values: {x} and {y}')
```

Answer:

Q10. *[Code Writing]*

Write a program that takes two numbers as input from the user and prints their sum, difference, product, and quotient (use descriptive labels for each).

Your Code:

03 Section 3: Arithmetic Operators

Q11. *[Calculation]*

Evaluate each expression manually and write the result:

```
print(17 // 5)    # Result: ___
print(17 %  5)    # Result: ___
print(2  ** 8)    # Result: ___
print(9  /  2)    # Result: ___
print(9  // 2)    # Result: ___
```

Answer:

Q12. *[Output Tracing]*

What will be the output of the following code?

```
a = 15
b = 4
print(a + b)
print(a - b)
print(a * b)
print(a / b)
print(a % b)
```

Answer:

Q13. *[Code Writing]*

Write a program that takes a number of seconds as input and converts it to hours, minutes, and remaining seconds. Use arithmetic operators only.

Your Code:

04 Section 4: Comparison & Logical Operators

Q14. *[Output Tracing]*

Write True or False for each expression:

```
x = 8
y = 12
print(x > y)          # ___
print(x < y)          # ___
print(x == 8)         # ___
print(x != y)         # ___
print(x >= 8 and y <= 15)  # ___
print(x > 10 or  y > 10)  # ___
print(not (x == y))       # ___
```

Answer:

Q15. *[Error Detection]*

Find the logical mistake in this code and correct it:

```
marks = 75
# Intention: pass if marks >= 50 AND marks <= 100
if marks >= 50 or marks <= 100:
    print('Pass')
else:
    print('Fail')
```

Mistake & Fix:

Q16. *[Calculation / Output Tracing]*

Evaluate and write the final output:

```
a = True
b = False
c = True
print(a and b)          # ___
print(a or  b)          # ___
print(not a)            # ___
print(a and b or  c)   # ___
print(not (a or  b))   # ___
```

Answer:

05 Section 5: Identity & Membership Operators

Q17. *[Output Tracing]*

What will be the output of the following code?

```
fruits = ['apple', 'banana', 'cherry']
print('apple'  in fruits)        # ___
print('grape'  in fruits)        # ___
print('mango' not in fruits)     # ___
```

Answer:

Q18. *[Output Tracing]*

Predict the output. Note the difference between **is** and **==**:

```
a = [1, 2, 3]
b = [1, 2, 3]
c = a
print(a == b)    # ___
print(a is b)    # ___
print(a is c)    # ___
```

Answer:

Q19. *[Fill in the Blank]*

Fill in the blank with **in**, **not in**, **is**, or **is not**:

a) To check if "e" exists in the string "hello": "e" ______ "hello"

b) To check if two variables point to the same object: x ______ y

c) To check "z" does NOT exist in "python": "z" ______ "python"

d) To check two variables do NOT share the same memory: a ______ b

Answer:

Q20. *[Mixed — Comprehensive]*

Write a program that:

1. Takes a word as input from the user.

2. Takes a sentence as input from the user.

3. Prints whether the word is found in the sentence using the **in** operator.

4. Prints the number of characters in the word using the **len()** function.

5. Prints whether the length of the word is greater than 5 using a comparison operator.

Your Code: